

Cost Analysis

2026-06-01

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1 Assumptions

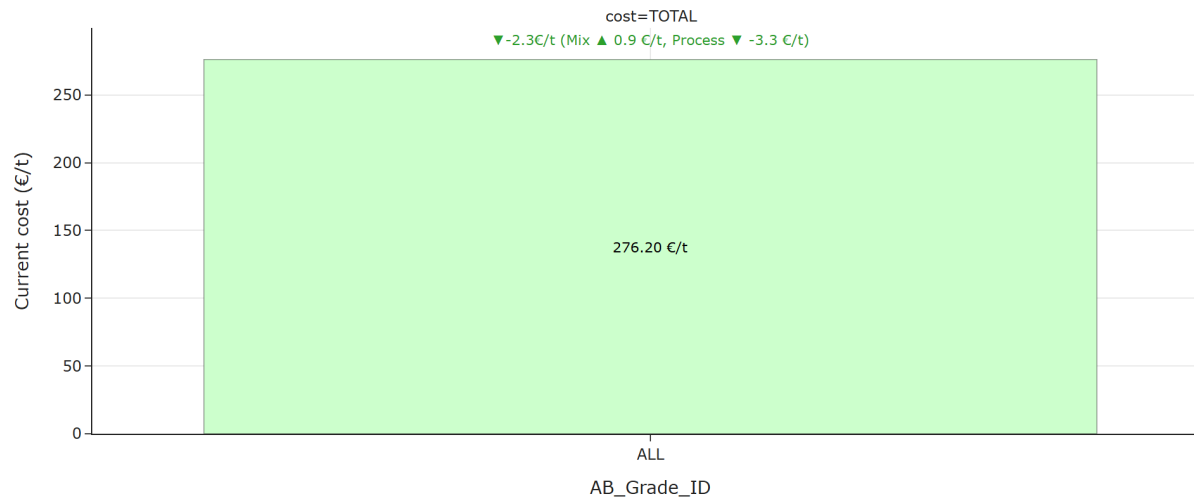
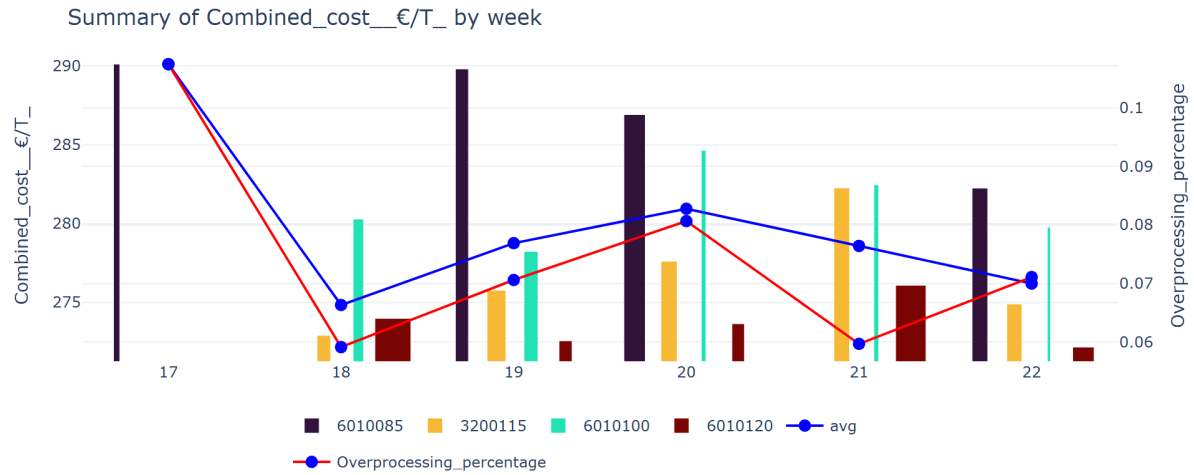
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** April 26, 2026 – June 01, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

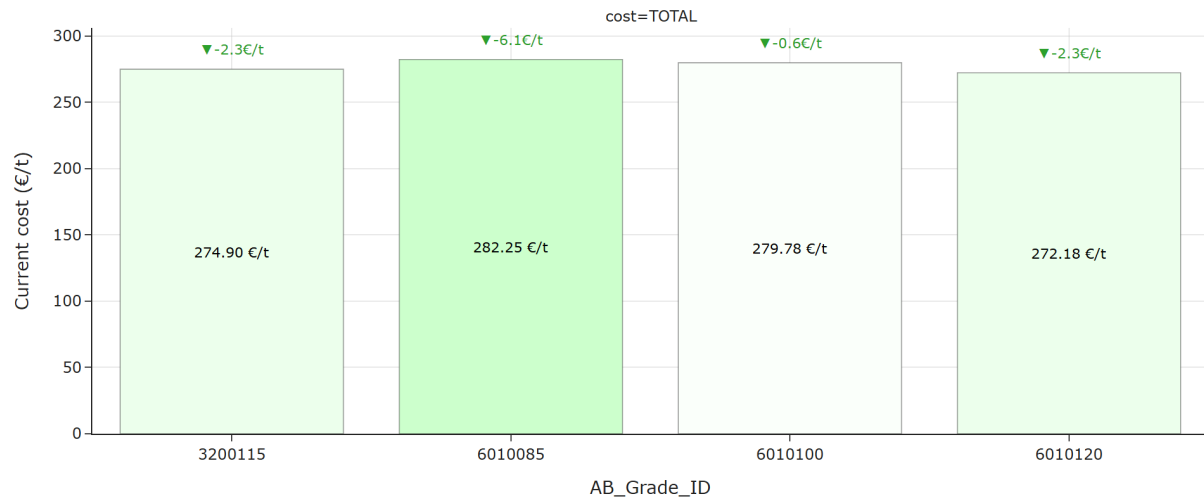
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 2.33 €/t, resulting in 276.20 €/t.

Breaking down the overall cost change, +0.93 €/t was driven by a more expensive mix of grades, and -3.26 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 2.31 €/t, resulting in 274.90 €/t.

For grade 6010085, TOTAL cost saw a decrease of 6.08 €/t, resulting in 282.25 €/t.

For grade 6010100, TOTAL cost saw a decrease of 0.56 €/t, resulting in 279.78 €/t.

For grade 6010120, TOTAL cost saw a decrease of 2.26 €/t, resulting in 272.18 €/t.

Looking across individual grades, the strongest cost decrease was observed for grade 6010085 (-6.08 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam_€/T** decreased on average (-3.95 €/t).
- **Starch_€/T** increased on average (+0.69 €/t).
- **Electricity_€/T** increased on average (+0.57 €/t).
- **Fibre_cost_€/T** decreased on average (-0.12 €/t).
- **Chemicals_€/T** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Starch_€/T** (+0.69 €/t on average), with the highest increase in grade 3200115 (+1.65 €/t) and the largest average decrease comes from **Steam_€/T** (-3.95 €/t on average), with the biggest drop in grade 6010085 (-6.85 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** decreased on average (-3.61 €/t).
- **Steam_Whitewater** decreased on average (-0.87 €/t).
- **Steam_Afterdryers** increased on average (+0.44 €/t).
- **Steam_Heatexchangers** increased on average (+0.10 €/t).
- **Steam_Starchkitchen** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Afterdryers** (+0.44 €/t on average), with the highest increase in grade 3200115 (+1.25 €/t) and the largest average decrease comes from **Steam_Predryers** (-3.61 €/t on average), with the biggest drop in grade 6010085 (-4.89 €/t). These components stand out and may require further investigation.



Average change by electricity:

- **Electricity_Other** increased on average (+0.40 €/t).
- **Electricity_Wiresection** increased on average (+0.12 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.04 €/t).
- **Electricity_Afterdryer** increased on average (+0.03 €/t).
- **Electricity_Freqconverters** increased on average (+0.03 €/t).
- **Electricity_Specific** increased on average (+0.03 €/t).
- **Electricity_Predryer** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.40 €/t on average), with the highest increase in grade 6010085 (+0.62 €/t) and the largest average decrease comes from **Electricity_Press_and_Steambox** (-0.04

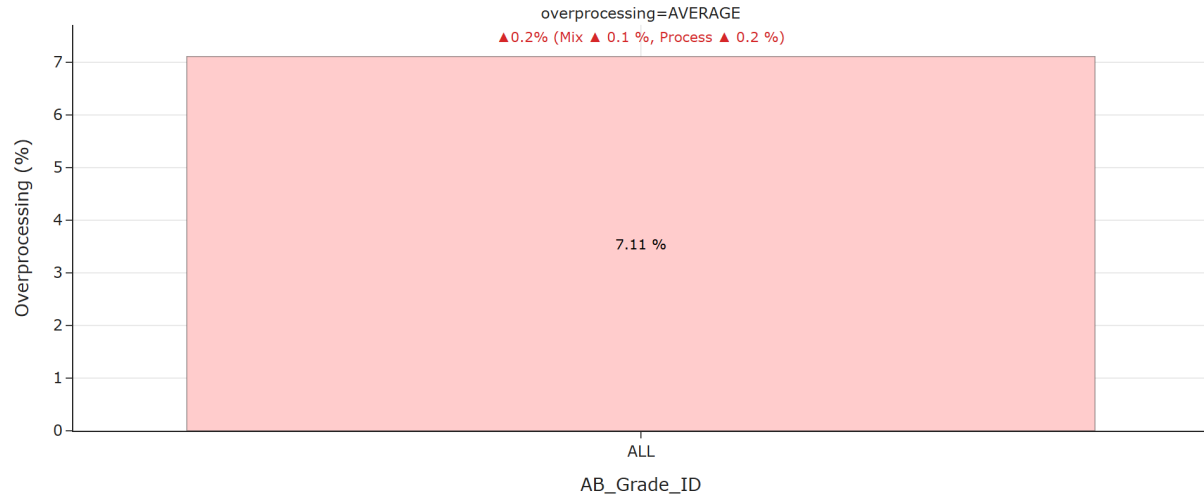
€/t on average), with the biggest drop in grade 6010085 (-0.05 €/t). These components stand out and may require further investigation.



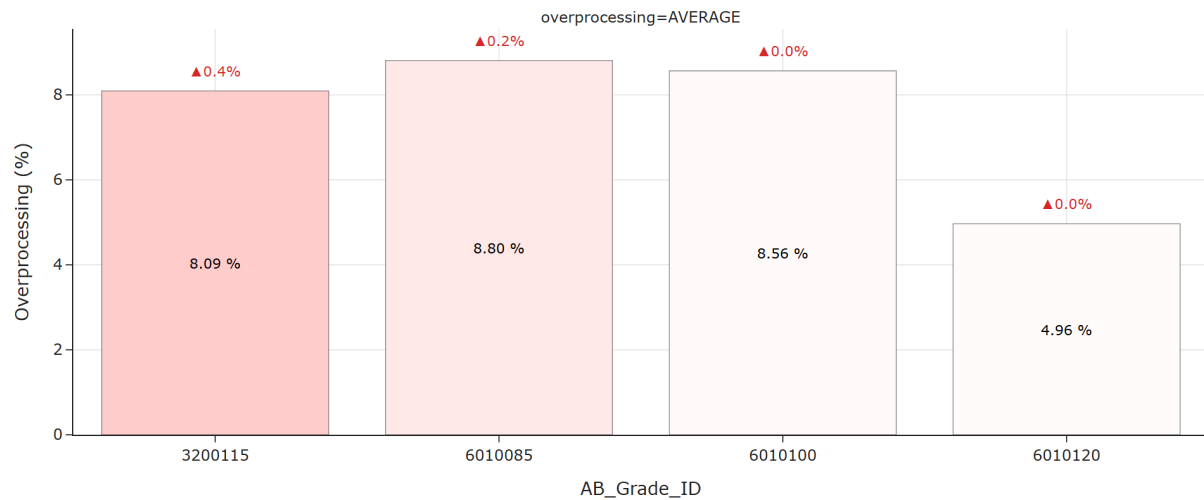
Average change by chemicals:

- **Sizing_Agent_€/T** increased on average (+0.16 €/t).
- **Defoamer_€/T** decreased on average (-0.03 €/t).
- **Retention_Aid_€/T** increased on average (+0.03 €/t).
- **Bentonite_2_€/T** decreased on average (-0.02 €/t).
- **Deaerator_€/T** increased on average (+0.01 €/t).
- **Fixative_2_€/T** decreased on average (-0.01 €/t).
- **Natriumhydroxide_€/T** decreased on average (-0.00 €/t).
- **Bentonite_1_€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/*T*** (+0.16 €/t on average), with the highest increase in grade 6010120 (+0.17 €/t) and the largest average decrease comes from **Defoamer_€/*T*** (-0.03 €/t on average), with the biggest drop in grade 6010120 (-0.05 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 0.25 %, resulting in 7.11 %.



For grade 3200115, TOTAL overprocessing saw an increase of 0.44 %, resulting in 8.09 %.

For grade 6010085, TOTAL overprocessing saw an increase of 0.20 %, resulting in 8.80 %.

For grade 6010100, TOTAL overprocessing saw an increase of 0.04 %, resulting in 8.56 %.

For grade 6010120, TOTAL overprocessing saw an increase of 0.03 %, resulting in 4.96 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade

3200115 (+0.44 %). These movements stand out and may require further investigation.



Average change by component:

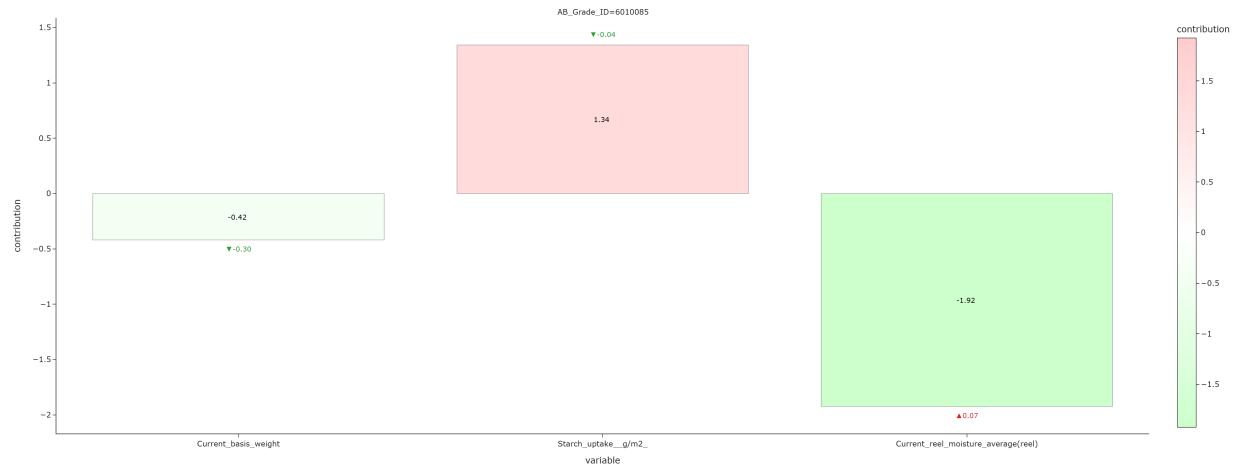
- **Burst** increased on average (+0.82 %).
- **SCT_CD** increased on average (+0.09 %).
- **CMT30** increased on average (+0.02 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+0.82 % on average), with the highest increase in grade 3200115 (+0.82 %). These components stand out and may require further investigation.

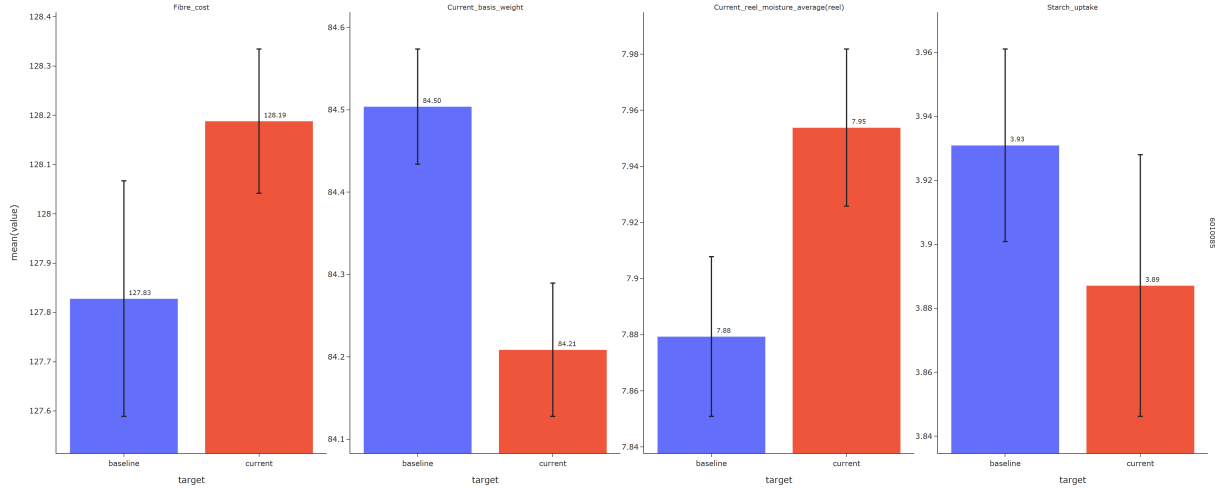
3 Breakdown by grades

3.1 6010085

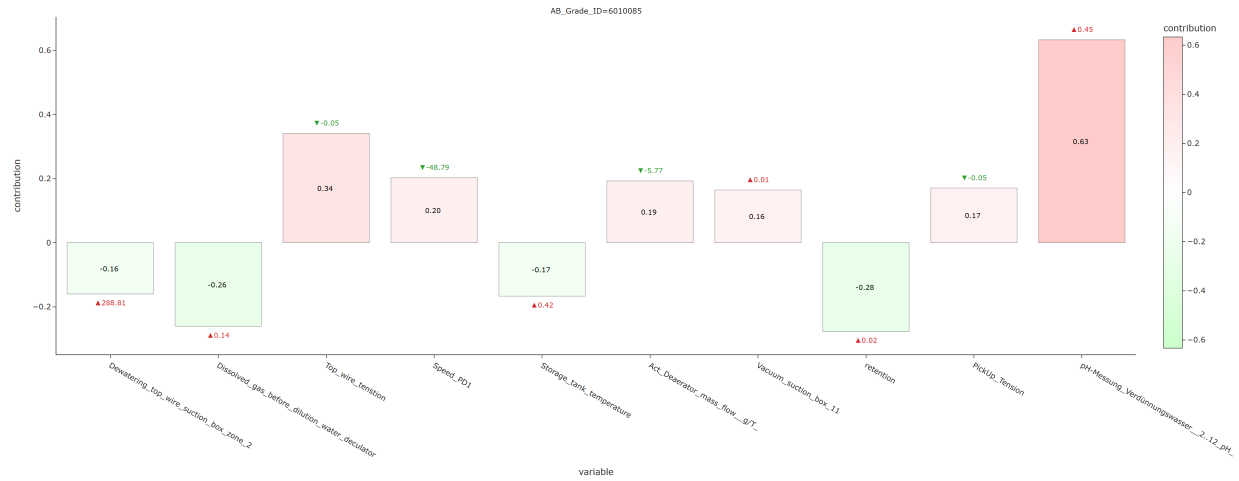
3.1.1 Fiber cost



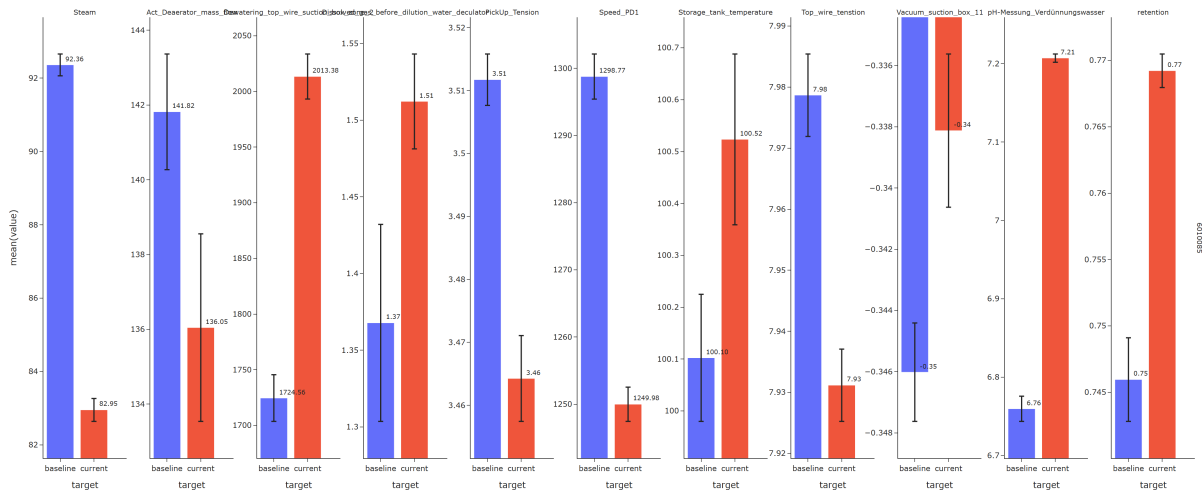
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Current reel moisture average(reel) increased (+0.07)



3.1.2 Steam cost



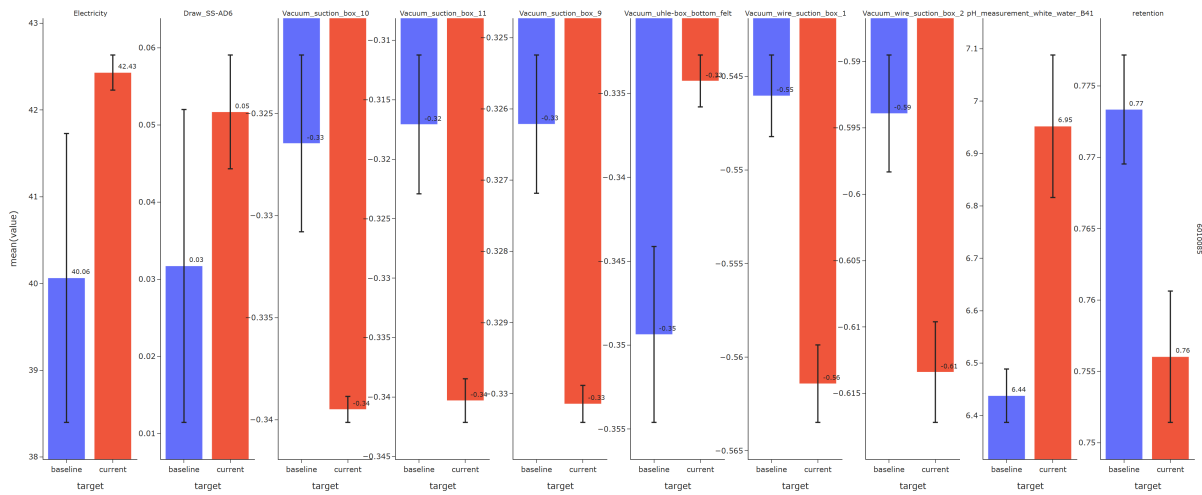
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Retention increased (+0.02) - Dissolved gas before dilution water decalator increased (+0.14) Cost-increasing drivers: - PH-Messung Verdünnungswasser 2..12 pH increased (+0.45) - Top wire tension decreased (-0.05) - Speed PD1 decreased (-48.79)



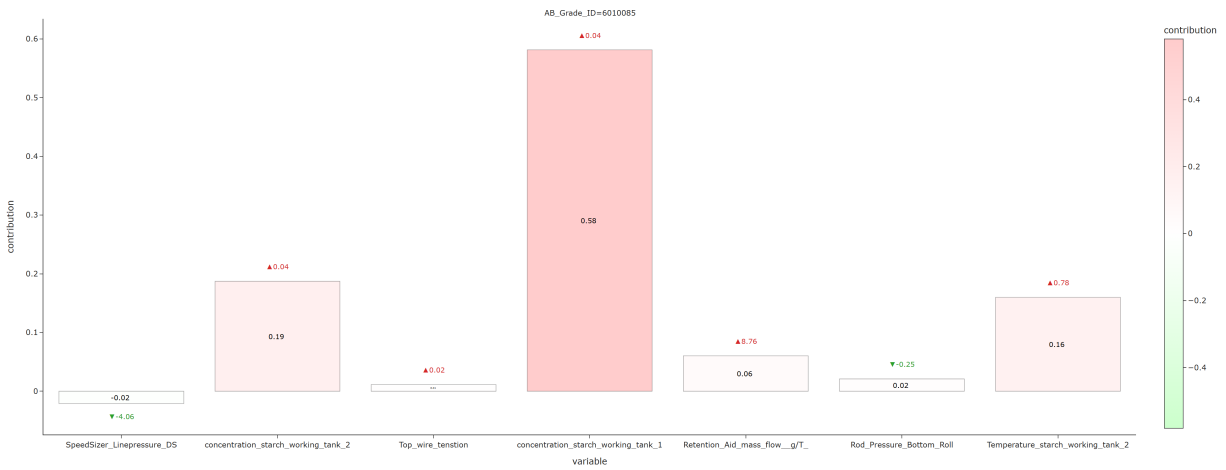
3.1.3 Electricity cost



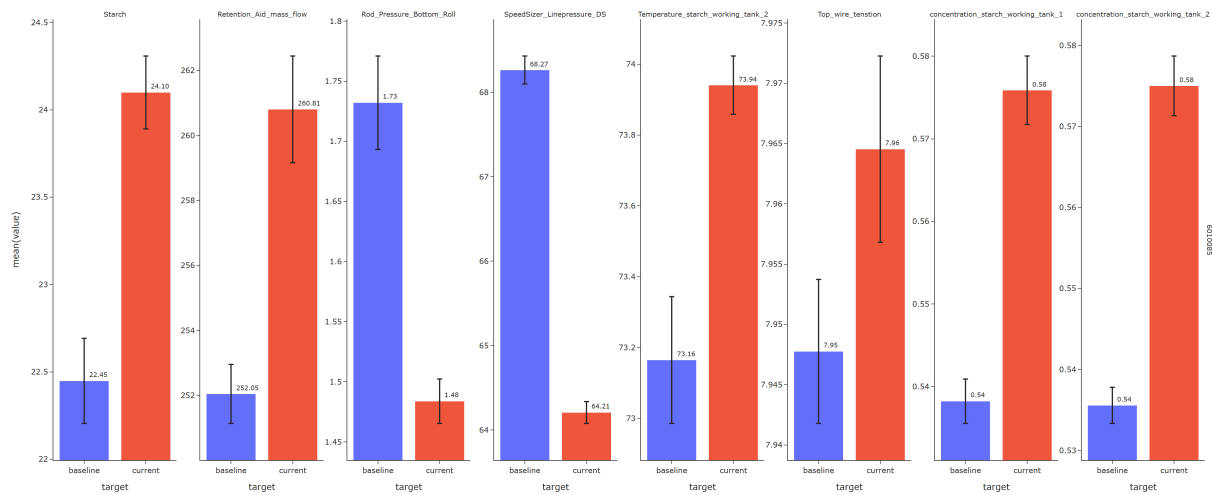
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 1 decreased (-0.02) Cost-increasing drivers: - Vacuum suction box 9 decreased (-0.00)



3.1.4 Starch cost



For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Concentration starch working tank 1 increased (+0.04) - Concentration starch working tank 2 increased (+0.04)

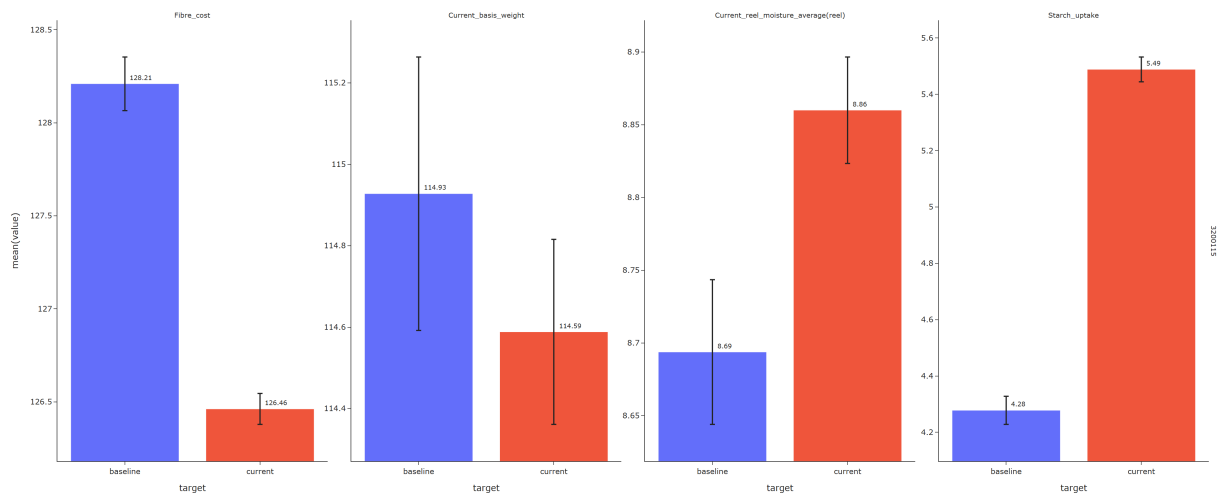


3.2 3200115

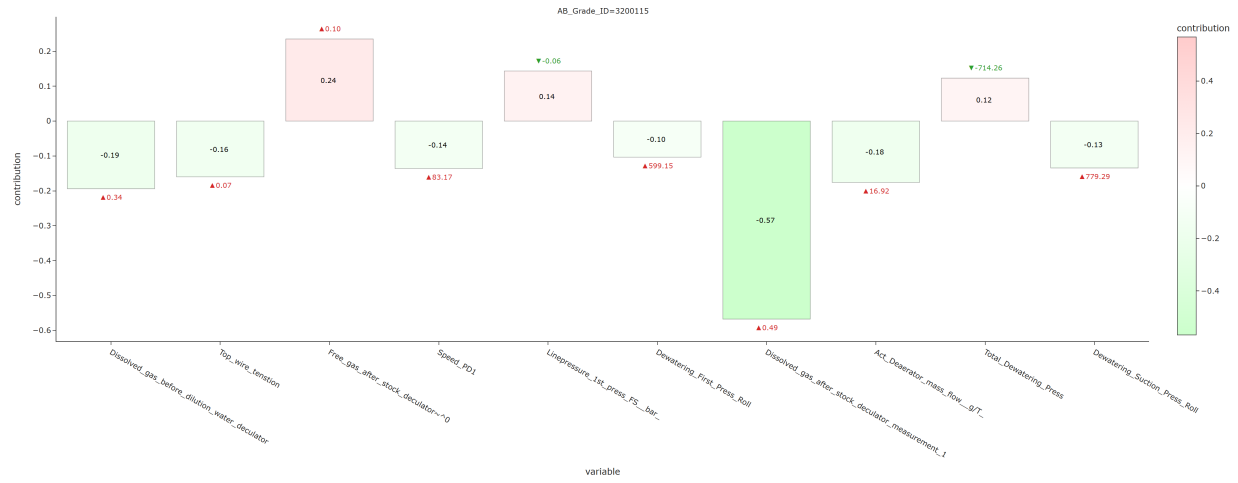
3.2.1 Fiber cost



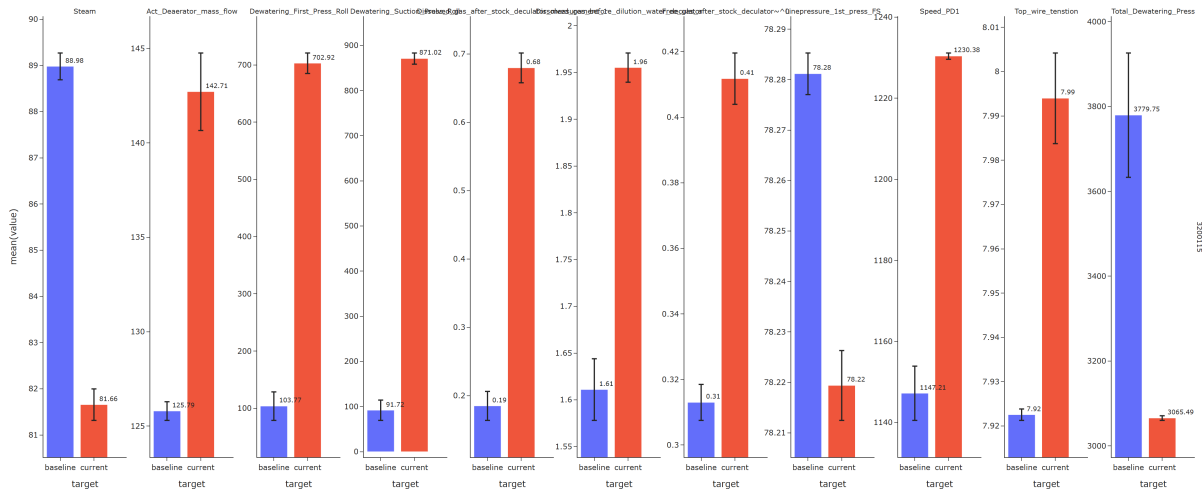
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+1.21)



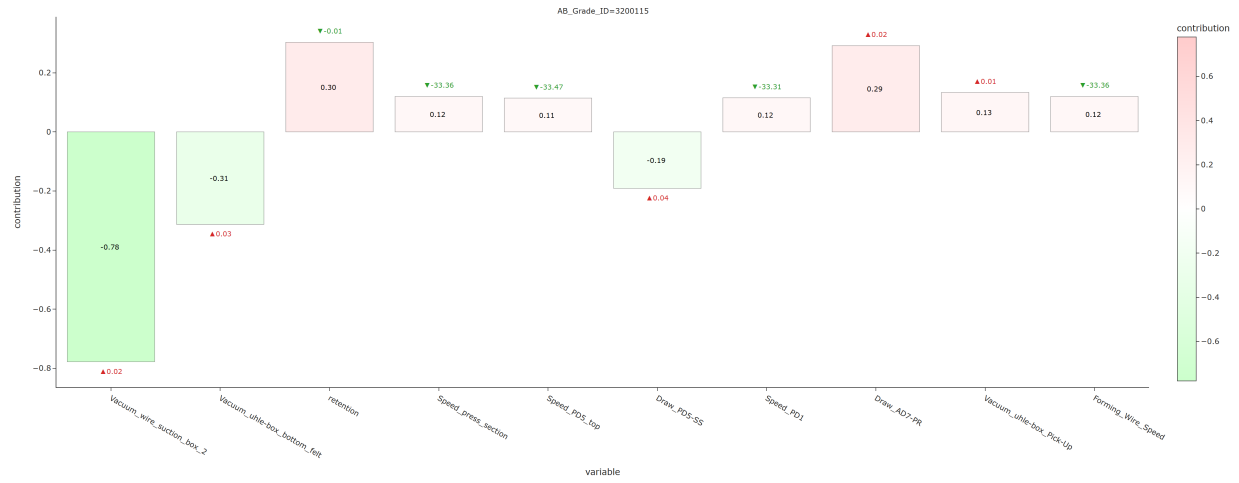
3.2.2 Steam cost



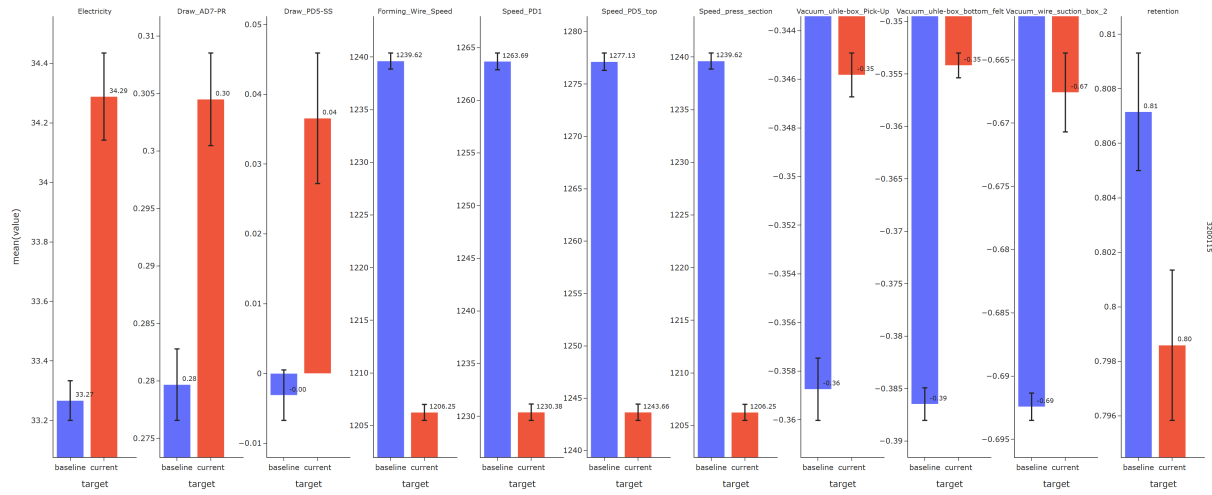
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas after stock deculator measurement 1 increased (+0.49) - Dissolved gas before dilution water deculator increased (+0.34) - Act Deaeator mass flow g/T increased (+16.92) - Top wire tenstion increased (+0.07) - Speed PD1 increased (+83.17) Cost-increasing drivers: - Free gas after stock decuator~0 increased (+0.10) - Linepressure 1st press FS bar decreased (-0.06)



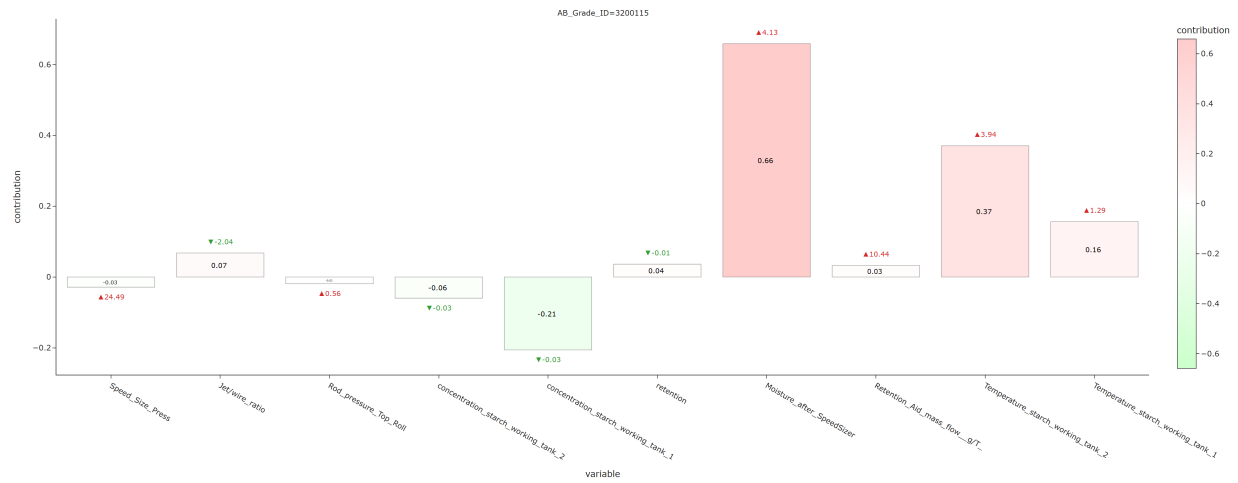
3.2.3 Electricity cost



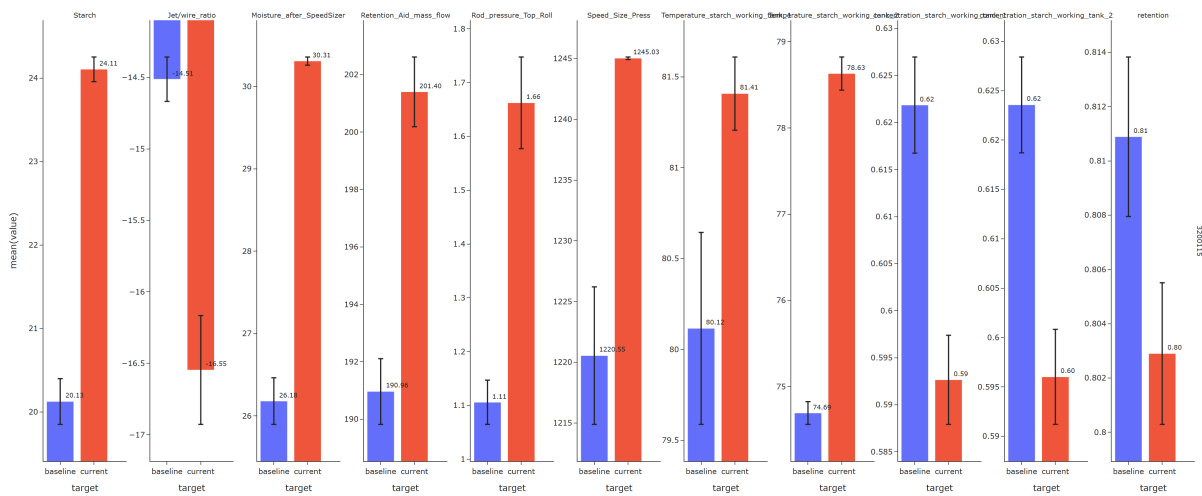
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.02) - Vacuum uhle-box bottom felt increased (+0.03) - Draw PD5-SS increased (+0.04) Cost-increasing drivers: - Retention decreased (-0.01) - Draw AD7-PR increased (+0.02)



3.2.4 Starch cost



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.03) Cost-increasing drivers: - Moisture after SpeedSizer increased (+4.13) - Temperature starch working tank 2 increased (+3.94)

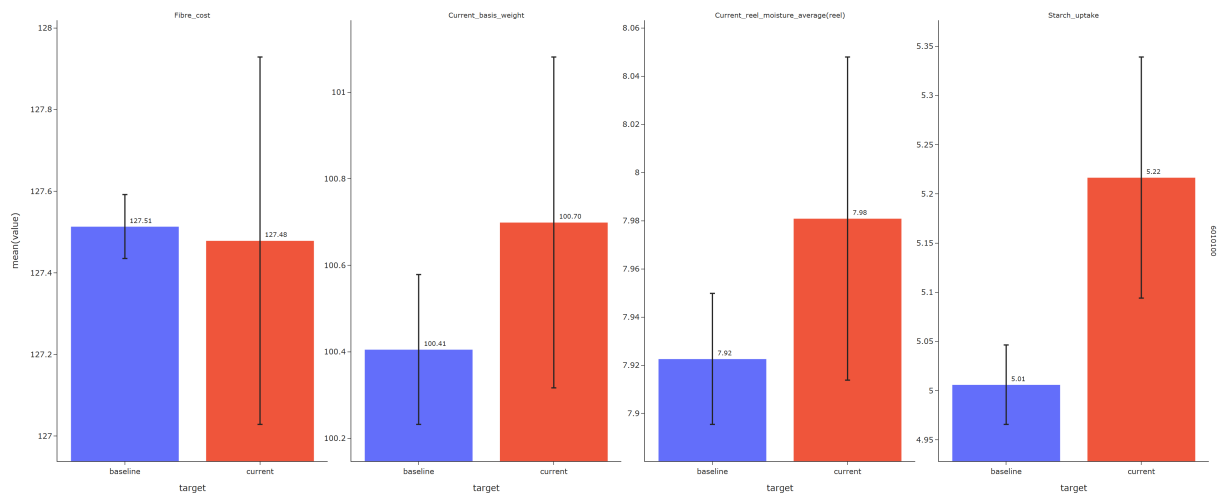


3.3 6010100

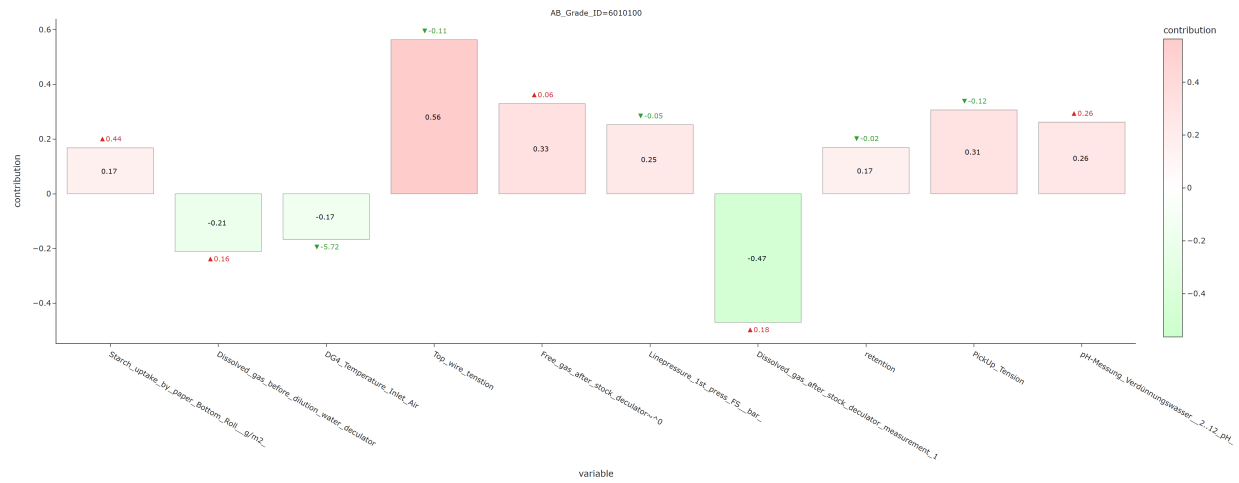
3.3.1 Fiber cost



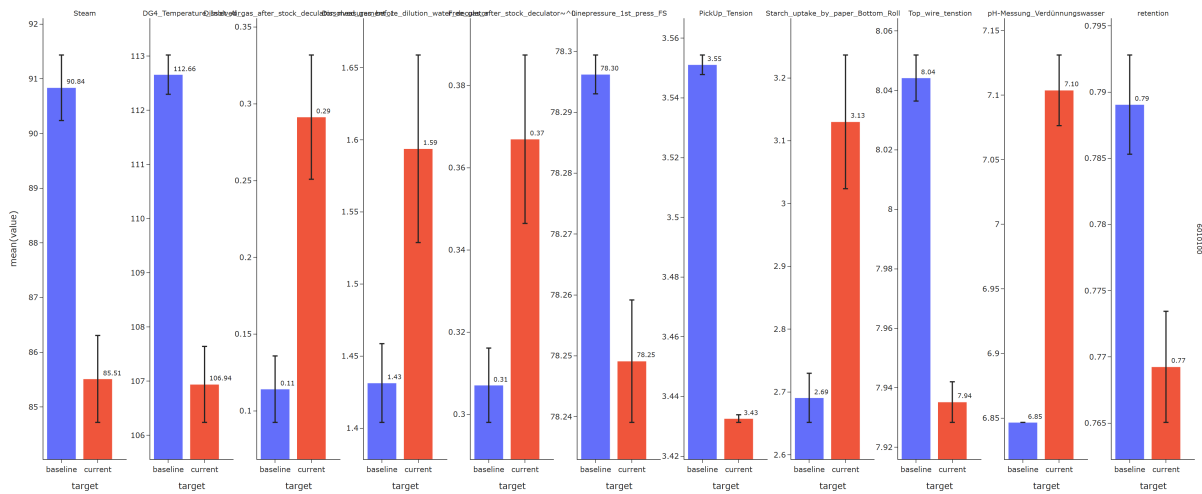
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.21)



3.3.2 Steam cost



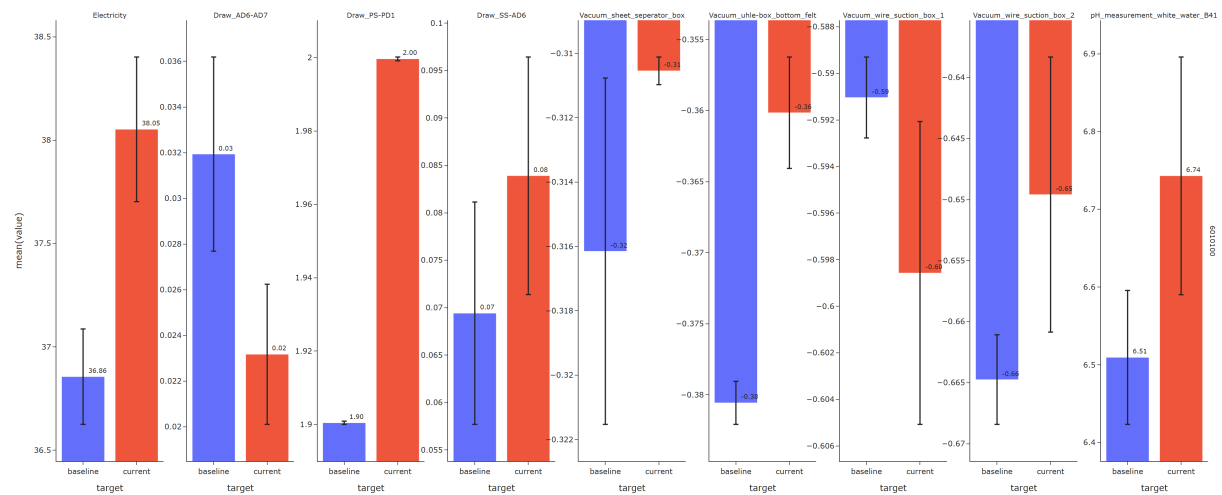
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas after stock deculator measurement 1 increased (+0.18) Cost-increasing drivers: - Top wire tension decreased (-0.11) - Free gas after stock deculator~0 increased (+0.06) - PickUp Tension decreased (-0.12) - PH-Messung Verdünnungswasser 2..12 pH increased (+0.26) - Linepressure 1st press FS bar decreased (-0.05)



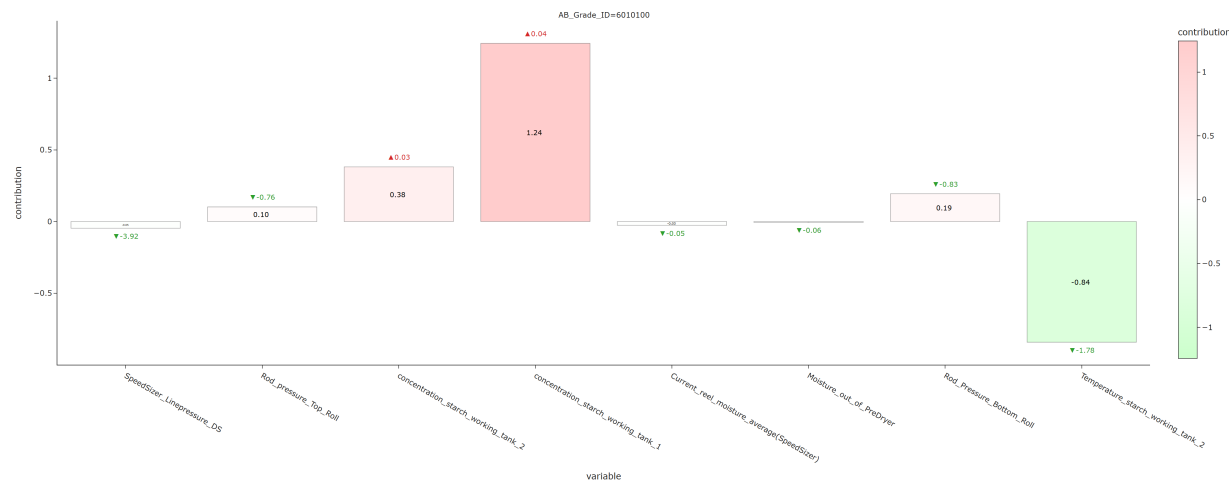
3.3.3 Electricity cost



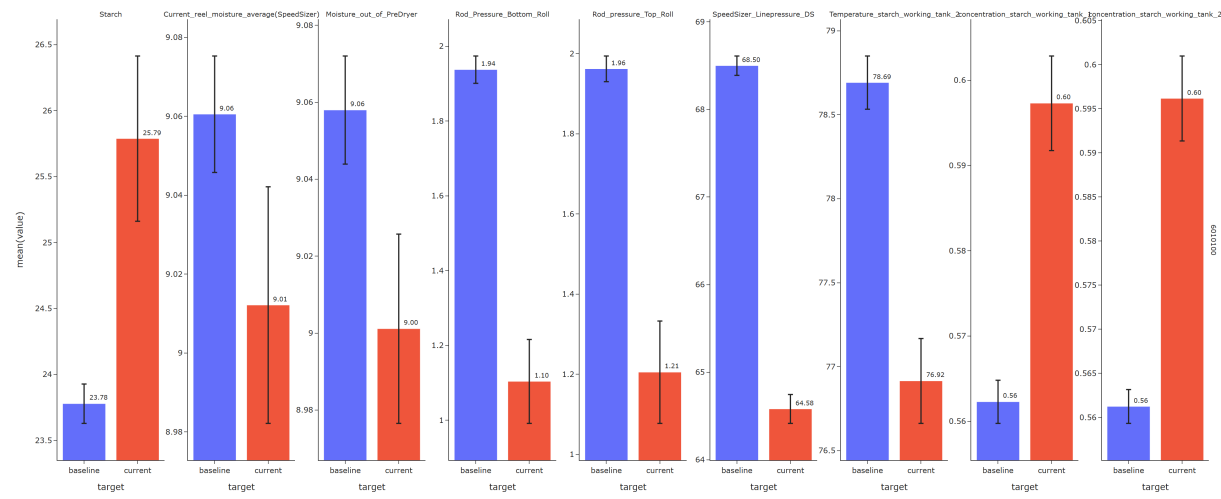
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.02) - Vacuum wire suction box 1 decreased (-0.01)



3.3.4 Starch cost



For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Temperature starch working tank 2 decreased (-1.78) Cost-increasing drivers: - Concentration starch working tank 1 increased (+0.04)

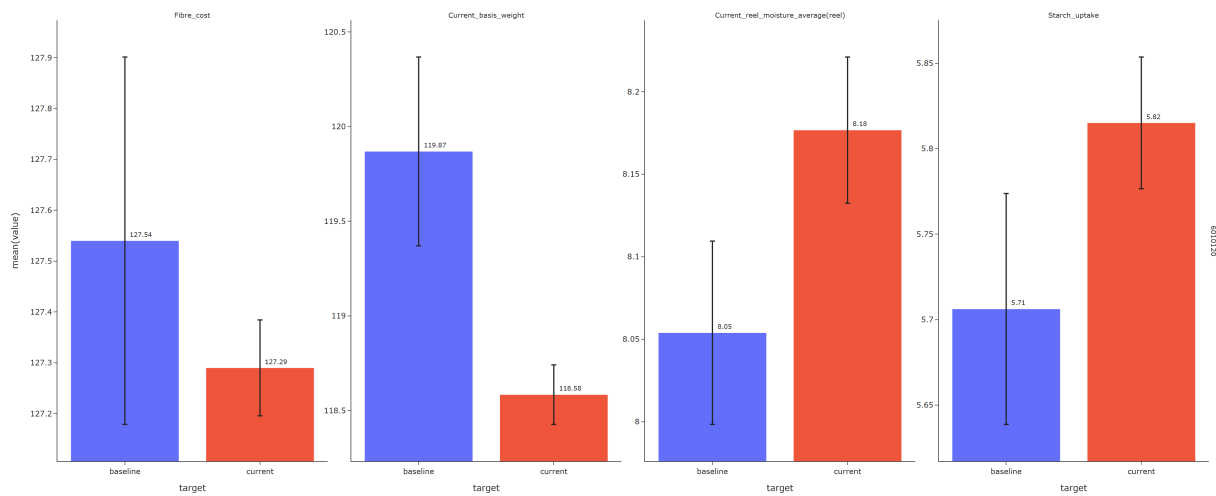


3.4 6010120

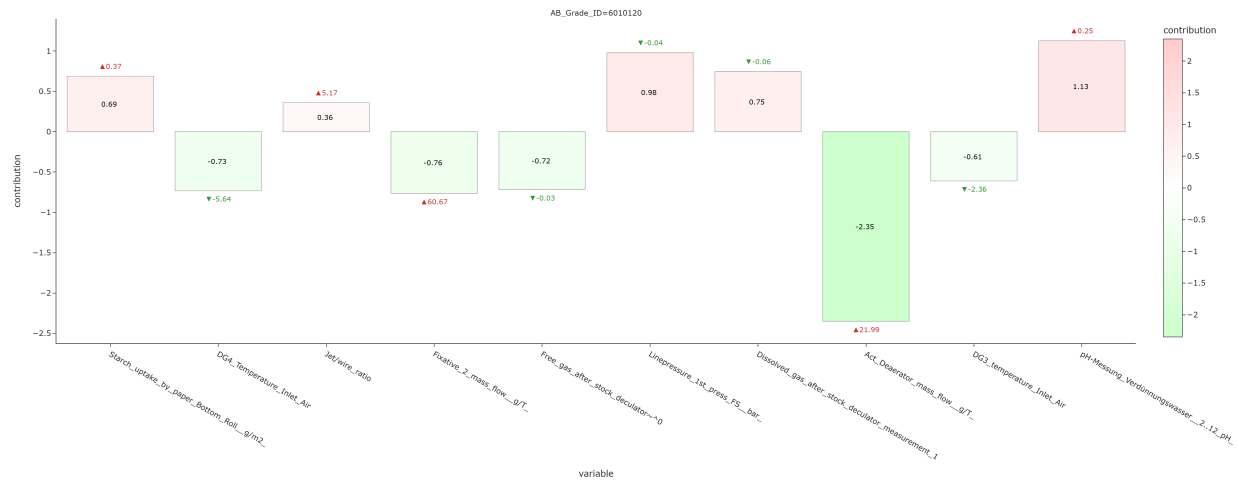
3.4.1 Fiber cost



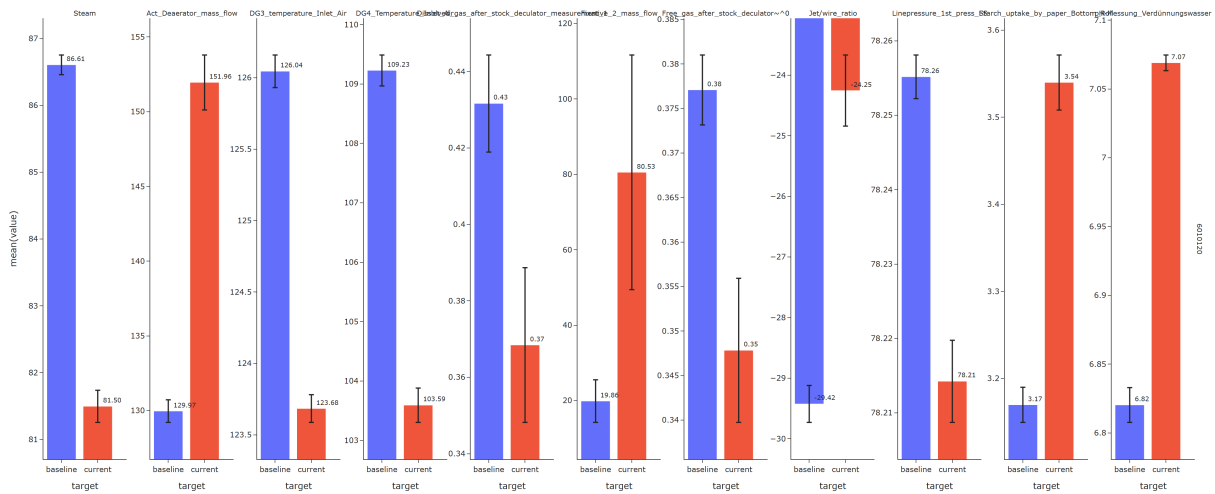
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Current reel moisture average(reel) increased (+0.12)



3.4.2 Steam cost



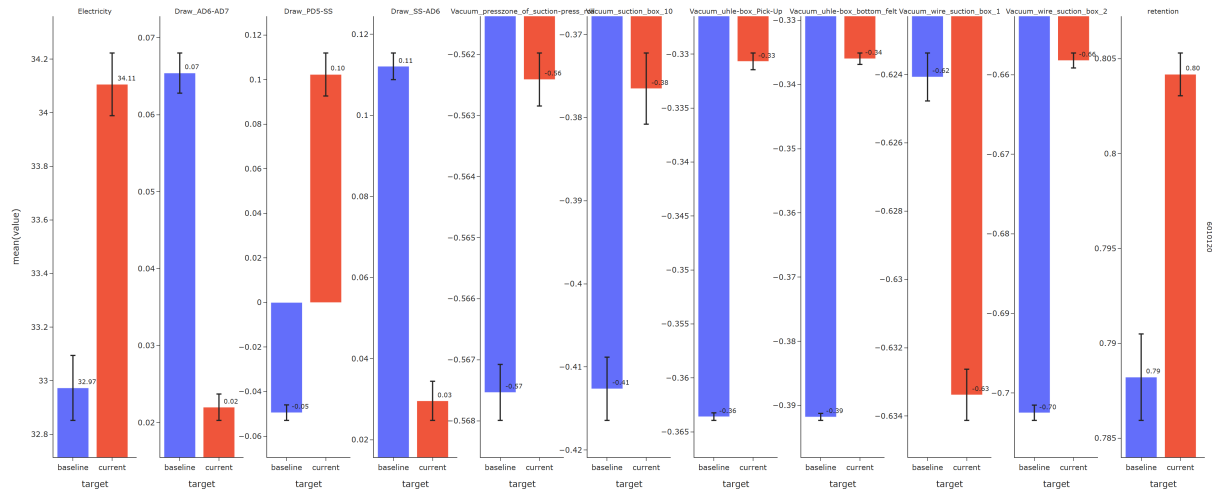
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Act Deaeator mass flow g/T increased (+21.99) - Fixative 2 mass flow g/T increased (+60.67) Cost-increasing drivers: - PH-Messung Verdünnungswasser 2..12 pH increased (+0.25) - Linepressure 1st press FS bar decreased (-0.04) - Dissolved gas after stock decuator measurement 1 decreased (-0.06)



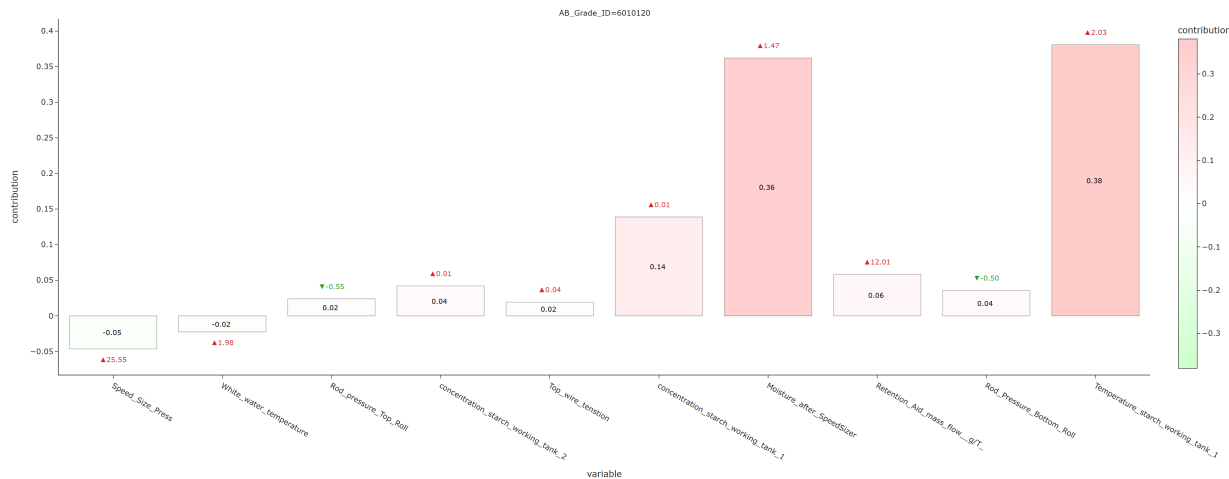
3.4.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.04) - Draw PD5-SS increased (+0.15) - Retention increased (+0.02) - Vacuum uhle-box bottom felt increased (+0.06) - Vacuum wire suction box 1 decreased (-0.01)



3.4.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Temperature starch working tank 1 increased (+2.03) - Moisture after SpeedSizer increased (+1.47) - Concentration starch working tank 1 increased (+0.01)

